

Number Toolkit

The words of number (factor, multiple, prime, square, cube), negatives and BIDMAS, and money problems with a clear decision at the end.



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THE TOPIC ON ONE MAP

Number Toolkit

01 Number families

a multiple of 8

a prime between 50 and 60

factors of 36

a cube number

02 Place value, negatives and BIDMAS

write in figures / words

7° below -3°

$-12 \div -2$

insert brackets

03 Money problems: decide

electricity bill

enough money?

which shop is cheaper?

cheapest packs

THE FIVE FAMILIES Know each word exactly:

ZONE 1 Factor

divides in:
factors of
12: 1, 2, 3, 4,
6, 12

ZONE 2 Multiple

in the times
table:
8, 16, 24, 32
...

ZONE 3 Prime

exactly two
factors:
2, 3, 5, 7, 11 ...

ZONE 4 Square

$n \times n$:
1, 4, 9, 16, 25
...

ZONE 5 Cube

$n \times n \times n$:
1, 8, 27, 64 ...

01 Number Families

Most “write down” marks are free — if you know the families by heart. Test each number in the list against the word.

WHEN YOU SEE → YOU DO

“a multiple of 7”

→ in the 7 times table: divide by 7, no remainder

“a prime number”

→ only 1 and itself divide it (1 is **not** prime)

“a square / cube number”

→ n^2 : 1, 4, 9, 16, 25, 36, 49, 64 ... n^3 : 1, 8, 27, 64, 125

“complete the list of factors”

→ work in pairs: 1×36 , 2×18 , 3×12 ...

“the difference between”

→ bigger – smaller

WORKED MODEL • FACTOR PAIRS

Complete the list of factors of 36: 1, 2, ..., 36.

1 1×36 , 2×18 , 3×12 , 4×9 , 6×6

factor pairs

ANSWER 1, 2, 3, 4, 6, 9, 12, 18, 36

× **TRAP** 1 is not a prime number (it has only one factor). 2 is the only even prime.

IDEA 01

Practise: Number Families

My questions first, then selected practice.

Q1 **DR ESLAM** Factor pairs

CH 1 • TOPIC 1 2 marks

Write down all the factors of 24.

Q2 **DR ESLAM** Which family?

CH 1 • TOPIC 1 3 marks

From 21, 23, 25, 27 write down (a) a prime number (b) a square number (c) a cube number.

SELECTED PRACTICE**Q3** **PRACTICE** Practice • Easy

CH 1 • TOPIC 1 1 mark

3 Give one reason why 0 is an even number.

(1 mark)

Q4

PRACTICE Practice · Easy

CH 1 · TOPIC 1 **3 marks**

7

32 33 34 35 36 37 38 39

From the list of numbers, write down

i) a multiple of 8,

[1]

ii) a square number,

[1]

- iii) a prime number.

[1]

(3 marks)

Q5

PRACTICE Practice · Easy

CH 1 • TOPIC 1 **2 marks**

15 Complete the list of factors of 36.

1, 2, , 36

(2 marks)

Q6**PRACTICE**

Practice • Medium

CH 1 • TOPIC 1 1 mark

6 Write down a square number greater than 10.

(1 mark)

Q7**PRACTICE**

Practice • Medium

CH 1 • TOPIC 1 2 marks

11 22 17 25 41 39 4

Work out the difference between the two prime numbers in the list above.

(2 marks)

Q8**PRACTICE**

Practice • Hard

CH 1 • TOPIC 1 1 mark

2 1, 2, 3, 5 and 7 are all common factors of two numbers.

Write down the digit that the two numbers must end in.

(1 mark)

02 Place Value, Negatives and BIDMAS

Temperatures are a number line: “below / fell / lower” move left, “above / rose / higher” move right. For a calculation, BIDMAS decides the order.

WHEN YOU SEE → YOU DO

“7°C below -3°C”, “fell by”

→ subtract: $-3 - 7 = -10$

“11 °C higher than”

→ add — or work back by subtracting

$-12 \div -2$

→ same signs → positive: 6

“write in figures”

→ use the place-value columns; fill empty places with 0

“insert one pair of brackets”

→ try each position; brackets first

Brackets

Indices

Divide / Multiply

Add / Subtract

WORKED MODEL • NUMBER LINE

The temperature at 07:00 is -3°C . This is 11°C higher than at 01:00. Find the temperature at 01:00.

1 01:00 is 11 degrees **lower**

reverse the direction

2 $-3 - 11$

move 11 left

ANSWER -14°C

× **TRAP** “Five million, two hundred and seven” is 5 000 207 — every empty column needs its 0.

IDEA 02

Practise: Place Value, Negatives and BIDMAS

My questions first, then selected practice.

Q9 DR ESAM Negatives

CH 1 • TOPIC 1 2 marks

Work out (a) $-4 - 7$ (b) $3 \times (-5) + 20$

SELECTED PRACTICE

Q10 PRACTICE Practice • Easy

CH 1 • TOPIC 1 1 mark

2 Circle the list of **all** the integers that satisfy $-2 < x \leq 4$

-2, -1, 0, 1, 2, 3	-1, 0, 1, 2, 3
-2, -1, 0, 1, 2, 3, 4	-1, 0, 1, 2, 3, 4

(1 mark)

Q11**PRACTICE**

Practice • Easy

CH 1 • TOPIC 1**1 mark**

- 6 At noon the temperature in Maseru was 21°C .
At midnight the temperature had fallen by 26°C .

Work out the temperature at midnight.

..... $^{\circ}\text{C}$

(1 mark)

Q12**PRACTICE**

Practice • Easy

CH 1 • TOPIC 1**1 mark**

- 9 Write down the temperature that is 7°C below -3°C .

..... $^{\circ}\text{C}$

(1 mark)

Q13**PRACTICE**

Practice · Easy

CH 1 • TOPIC 1 1 mark

13 Write the number five million, two hundred and seven in figures.

(1 mark)

Q14**PRACTICE**

Practice · Medium

CH 1 • TOPIC 1 1 mark

7 The lowest temperature recorded at Scott Base in Antarctica is -57.0°C .
The highest temperature recorded at Scott Base is 63.8°C more than this.

What is the highest temperature recorded at Scott Base?

..... $^{\circ}\text{C}$

(1 mark)

Q15**PRACTICE**

Practice · Hard

CH 1 • TOPIC 1 1 mark

5 Insert one pair of brackets to make this calculation correct.

$$7 - 5 - 3 + 4 = 9$$

(1 mark)

03 Money Problems: Decide

Work out each option separately, write money with 2 decimal places, then **answer the question** in a sentence: yes / no / which one, with the numbers.

WHEN YOU SEE → YOU DO

“Does he have enough money?”

→ total cost, compare with the money, say yes/no

“Which is cheaper? You must show all your working”

→ cost of each option, then choose

“plus VAT at 20%”

→ $\times 1.2$

“at least 100 tiles”, packs

→ try the combinations; the cheapest wins

price in pence, answer in pounds

→ $900\text{p} = \text{£}9.00$

Each option

Same units

Compare

Sentence answer

WORKED MODEL • WHICH SHOP?

25 plants: £2.39 each, or a pack of 25 for £52.50 plus VAT at 20%.

1 $25 \times 2.39 = \text{£}59.75$

option 1

2 $52.50 \times 1.2 = \text{£}63.00$

option 2

ANSWER Kirsty's Plants: $\text{£}59.75 < \text{£}63.00$

× **TRAP** A number alone is not an answer to “should he ...?”. Write the decision and the two amounts.

IDEA 03

Practise: Money Problems: Decide

My questions first, then selected practice.

Q16**DR ESLAM**

Better value

CH 1 • TOPIC 1

3 marks

Water: one bottle £1.35, or a pack of 6 bottles for £7.50. Which is better value? Show your working.

SELECTED PRACTICE

Q17**PRACTICE**

Practice • Easy

CH 1 • TOPIC 1 4 marks

- 1 Here is part of Gary's electricity bill.

Electricity bill	
New reading	7155 units
Old reading	7095 units
Price per unit 15p	

Work out, in pounds, how much Gary has to pay for the units of electricity he used.

(4 marks)

Q18**PRACTICE**

Practice • Medium

CH 1 • TOPIC 1 6 marks

- 1 (a) The diagram shows a patio in the shape of a rectangle.

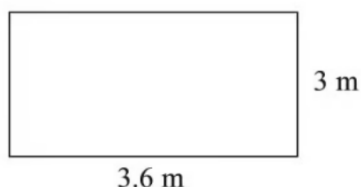


Diagram **NOT**
accurately drawn

The patio is 3.6m long and 3 m wide.

Matthew is going to cover the patio with paving slabs.
Each paving slab is a square of side 60 cm.

Matthew buys 32 of the paving slabs.

Does Matthew buy enough paving slabs to cover the patio?
You must show all your working.

(3 marks)

- (b) The paving slabs cost £8.63 each.

Work out the total cost of the 32 paving slabs.

(3 marks)

Q19**PRACTICE**

Practice • Medium

CH 1 • TOPIC 1 4 marks

2 Steve wants to put a hedge along one side of his garden.

He needs to buy 27 plants for the hedge. Each plant costs £5.54.

Steve has £150 to spend on plants for the hedge.

Does Steve have enough money to buy all the plants he needs?

(4 marks)

Q20**PRACTICE**

Practice • Medium

CH 1 • TOPIC 1 5 marks

4 Tom is going to buy 25 plants to make a hedge.

Here is information about the cost of buying the plants.

Kirsty's Plants

£2.39 each

Hedge World

Pack of 25

£52.50 plus VAT at 20%

Tom wants to buy the 25 plants as cheaply as possible.

Should Tom buy the plants from Kirsty's Plants or from Hedge World?

You must show all your working.

(5 marks)

Q21**PRACTICE**

Practice • Hard

CH 1 • TOPIC 1 4 marks**3 (a)** Tom is tiling a wall.

He needs to buy at least 100 tiles.

The tiles are sold in large packs and small packs.

Large pack 40 tiles £18

Small pack 28 tiles £14

Special offer

25% reduction when you buy 3 or more **large** packs

Work out the cheapest cost for Tom to buy the packs of tiles he needs.

£.....

(3 marks)**(b)** Tom is also tiling a floor.

The floor is a rectangle with length 600 cm and width 240 cm

Each tile is a square with side 40 cm

Tom uses this method to work out the number of tiles he needs.

Number of tiles that will fit along the length = $600 \div 40$
= 15Number of tiles that will fit along the width = $240 \div 40$
= 6Total number of tiles needed = $15 + 6$
= 21

Give a reason why Tom's method is wrong.

(1 mark)

Q22**PRACTICE**

Practice • Hard

CH 1 • TOPIC 1 4 marks

- 4 The table shows the children nominated to win the subject prize in Mathematics and the subject prize in English.

Mathematics	English
Alice	Alice
Ben	Claire
Emma	Gabi
Paddy	Simon

The winner of each subject prize is picked at random.
It is possible for Alice to win both prizes.

In what percentage of the combinations of prize winners does Alice win **at least** one prize?

(4 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

ZONE 1 Factor divides in	ZONE 2 Multiple times table	ZONE 3 Prime two factors	ZONE 4 Square n^2	ZONE 5 Cube n^3
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- 01 Families** Factor pairs; 1 is not prime; know squares to 225 and cubes to 125.
- 02 Negatives** Below/fell → left; above/rose → right. Same signs $\times \div$ → positive. BIDMAS.
- 03 Money** Each option, same units, 2 d.p., then a sentence with the decision.

BOTTOM LINE Know the words → draw the number line → money: compare, then decide in words.

Answers

Number Toolkit • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • NUMBER FAMILIES

Q1	Dr Eslam • Factor pairs	1, 2, 3, 4, 6, 8, 12, 24
Q2	Dr Eslam • Which family?	(a) 23 (b) 25 (c) 27
Q3	Practice • Easy	it is a multiple of 2 (divides by 2 exactly)
Q4	Practice • Easy	(i) 32 (ii) 36 (iii) 37
Q5	Practice • Easy	3, 4, 6, 9, 12, 18
Q6	Practice • Medium	16 (or 25, 36, ...)
Q7	Practice • Medium	$41 - 17 = 24$
Q8	Practice • Hard	0

IDEA 02 • PLACE VALUE, NEGATIVES AND BIDMAS

Q9	Dr Eslam • Negatives	(a) -11 (b) 5
Q10	Practice • Easy	-1, 0, 1, 2, 3, 4
Q11	Practice • Easy	-5
Q12	Practice • Easy	-10
Q13	Practice • Easy	5 000 207
Q14	Practice • Medium	6.8
Q15	Practice • Hard	$7 - (5 - 3) + 4 = 9$

IDEA 03 • MONEY PROBLEMS: DECIDE

Q16 Dr Eslam • Better value

pack: $7.50 \div 6 = £1.25 < £1.35$ — the pack

Q17 Practice • Easy

$$60 \times 15 = 900\text{p} = £9.00$$

Q18 Practice • Medium

(a) $6 \times 5 = 30$ slabs needed: yes (b) £276.16

Q19 Practice • Medium

$$27 \times 5.54 = £149.58: \text{yes, } 42\text{p left}$$

Q20 Practice • Medium

Kirsty £59.75, Hedge World £63.00: Kirsty's

Q21 Practice • Hard

(a) 3 large packs: £40.50 (b) multiply, not add: $15 \times 6 = 90$

Q22 Practice • Hard

$$\frac{7}{16} = 43.75\%$$